

Assignment no. 2

Generative AI

GAI4UE

SS 2026

In this assignment you will compare two paradigms for **out-of-distribution (OOD) detection**. The model is trained on UC Merced (aerial imagery, in-distribution). At test time, FIDS30 fruit images serve as OOD samples. In both approaches the threshold is set at the 95th percentile of the in-distribution score so that the false alarm rate is fixed at $\sim 5\%$.

Exercise 1 (4 points) Prepare the two datasets:

- Load UC Merced from HuggingFace ([blanchon/UC_Merced](#)) and split it 70% / 15% / 15% (stratified).
- Download and extract FIDS30 (<https://data.vicos.si/datasets/FIDS30/FIDS30.zip>) into PrepData/FIDS30/.
- Visualize a few samples from each dataset side by side.

Exercise 2 (8 points) Train a **deep ensemble** of 10 EfficientNet-B0 models on UC Merced (5 epochs each, different seed per model):

- Run inference with all 10 models on UC Merced test **and** FIDS30, collect softmax outputs.
- Compute the per-sample **variance across models** (averaged over classes) as OOD score.
- Threshold at the 95th percentile of in-distribution variance; report detection rate and false alarm rate; plot histograms with the threshold line.

Exercise 3 (8 points) Train a **convolutional autoencoder** on UC Merced:

- Use MSE loss and an Adam optimizer ($lr = 10^{-3}$), 15 epochs.
- Compute the per-sample **reconstruction MSE** on UC Merced test and FIDS30.
- Threshold at the 95th percentile of in-distribution MSE; report detection rate and false alarm rate; plot histograms with the threshold line.
- Visualize input / reconstruction pairs for both datasets.

Skeleton: A starter repository with notebook templates is available at:
https://github.com/MustafaArikan/GAI4_course (folder HW_2).

Deliverables:

- 3 Jupyter notebooks exported as HTML
- 1-page report (PDF): compare the two methods (detection rate, false alarm rate, training cost, need for labels) and briefly discuss which wins and why.

Submission: electronically via Moodle; **Deadline: Thu, May 7, 2026, 23:59**